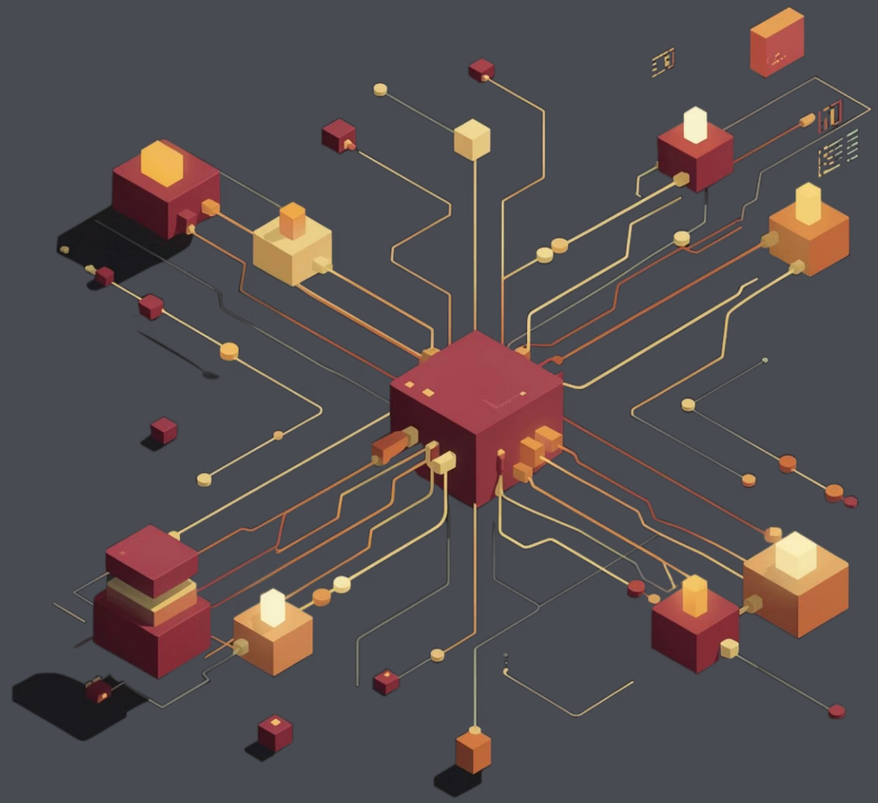
AI-Based Anomaly Detection in Network Security

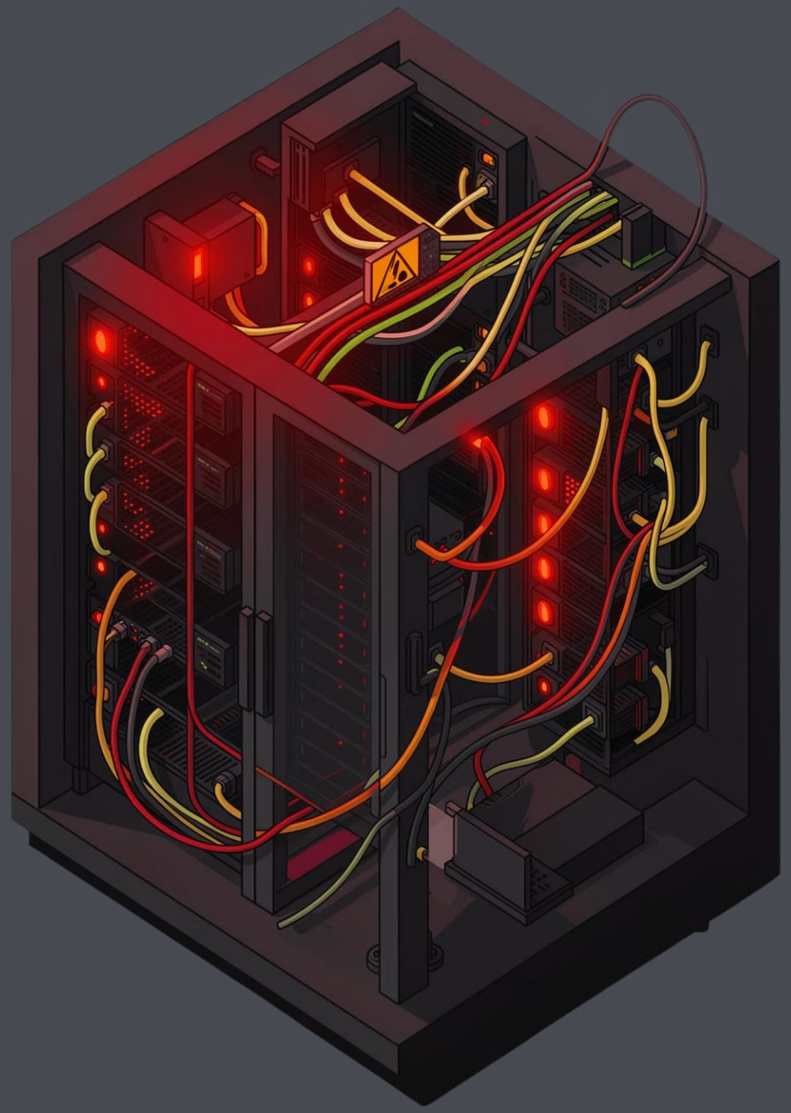
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ALP KALE

EGE KAYA

DOGAN EGE BULTE





The Problem with Signature-Based Detection

Traditional systems like Snort and Suricata match traffic against known attack patterns. They are fast and precise — but fundamentally reactive.

Zero-Day Blind Spot

New attacks spread undetected until rules are written.

Evasion by Design

Single-byte changes, encryption, or session fragmentation invalidate rules.

ICMP Tunnelling

Data hidden in ping packets produces no recognizable signature.

Signature-Based Detection Limitations

Signature-based detection systems, while efficient for known threats, exhibit inherent limitations that leave networks vulnerable to sophisticated and evolving attacks. These limitations stem from their reliance on pre-defined patterns and their reactive nature.



Zero-Day Blind Spot

New attacks spread undetected until rules are written, leaving a critical window of vulnerability for previously unseen threats.



Evasion by Design

Single-byte changes, encryption, or session fragmentation invalidate rules, allowing attackers to easily bypass detection mechanisms.

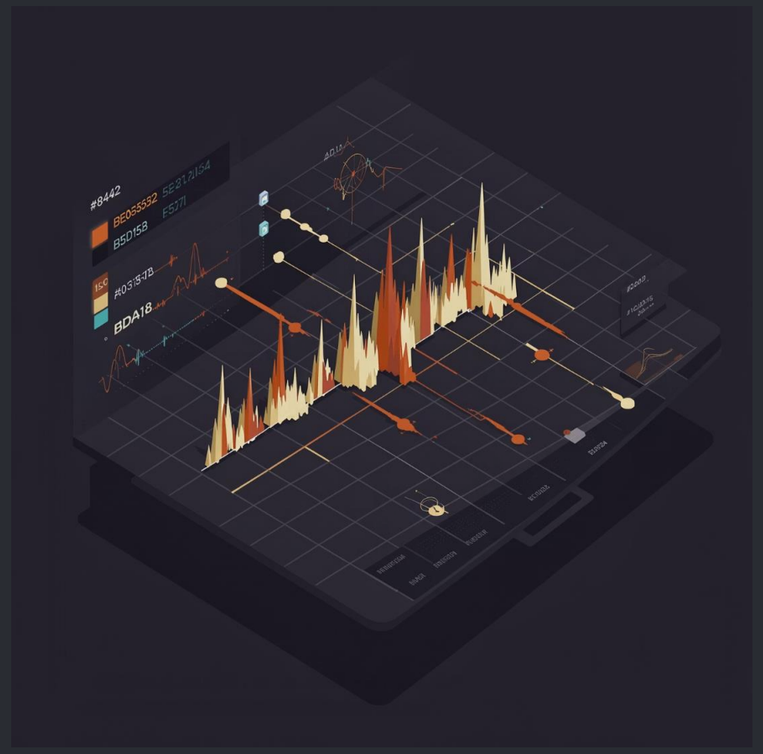


ICMP Tunnelling

Data hidden in ping packets produces no recognizable signature, making it a stealthy method for exfiltration and command & control.

These fundamental weaknesses highlight the need for more adaptive and proactive security measures.

Types of Network Anomalies



Unusual Traffic Volume

DDoS floods or sharp drops signaling host compromise.

Data Exfiltration

DNS/ICMP tunnelling moves stolen data without triggering alarms.

Unauthorized Access

Brute-force, SQL injection, zero-day exploits absent from rule databases.

Network Reconnaissance

Port scans and host sweeps with distinctive timing patterns.

Anomaly Detection Fundamentals

Unlike signature-based systems that rely on known attack patterns, AI-based anomaly detection (AI-ADS) operates on a fundamentally different principle: understanding and profiling "normal" behavior to identify "abnormal" deviations. This allows for the detection of novel and evolving threats.

Normal Network Behavior is Learnable

AI-ADS establishes a baseline of expected network activities, identifying patterns in traffic flow, user actions, and system logs.

AI-ADS Trains on Normal Traffic

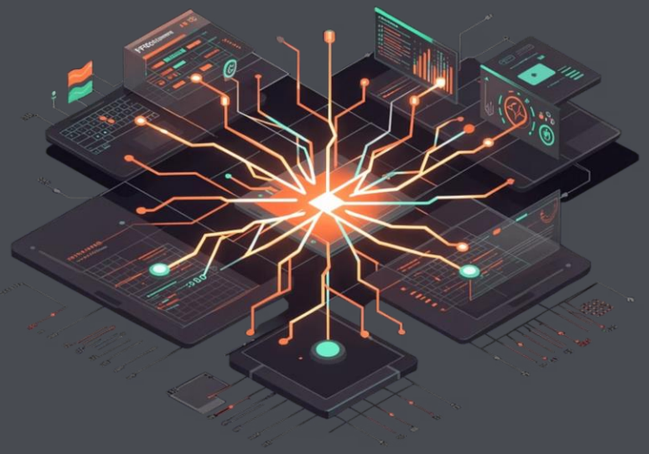
Machine learning models are fed extensive datasets of legitimate network operations to build a comprehensive understanding of what constitutes "normal."

Flags Deviations with an Anomaly Score

Any activity that significantly deviates from the learned normal baseline is flagged, assigned an anomaly score, and potentially identified as a threat.

⚠ Key Trade-off

Tight models catch subtle attacks but raise false positives; loose models reduce noise but may miss borderline threats.



AI-Based Anomaly Detection: Core Principle

Normal network behaviour is **learnable**. An AI-ADS trains on normal traffic, then flags deviations with an anomaly score.



Key trade-off: tight models catch subtle attacks but raise false positives; loose models reduce noise but may miss borderline threats.

Classical Machine Learning Methods



Isolation Forest

Unsupervised ensemble. Anomalies isolate in fewer partitions. Linear complexity — ideal for high-speed monitoring with no labelled data.



One-Class SVM

Finds the smallest boundary around normal flows. RBF kernel models non-linear shapes. Limited scalability on large datasets.



Random Forest

Supervised ensemble with feature importance scores. Strong performance but depends on labelled, up-to-date attack data.

Machine Learning Algorithms Overview

Here's a detailed comparison of classical machine learning methods used in anomaly detection, highlighting their core principles, advantages, and limitations:



Isolation Forest

Unsupervised ensemble. Anomalies isolate in fewer partitions.

Advantages: Linear complexity — ideal for high-speed monitoring with no labelled data.



One-Class SVM

Finds the smallest boundary around normal flows. RBF kernel models non-linear shapes.

Limitations: Limited scalability on large datasets.



Random Forest

Supervised ensemble with feature importance scores.

Performance: Strong performance but depends on labelled, up-to-date attack data.

Deep Learning Approaches

Autoencoders

Compress and reconstruct normal traffic. High reconstruction error = anomaly. **Kitsune** uses an ensemble for real-time detection.

LSTM Networks

Models sequential flow patterns over time. Detects slow-and-low exfiltration and APTs invisible in single-flow analysis.

CNNs

Treats packet windows as images. Hybrid CNN-LSTM architectures achieve top benchmark scores on NSL-KDD and CICIDS-2017.

Deep Learning Models for Detection



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Methodology Comparison

Method	Supervision	Scalability	Temporal	Best For
Isolation Forest	Unsupervised	High	No	Volumetric anomalies
OC-SVM	Unsupervised	Low	No	Compact normal clusters
Random Forest	Supervised	High	No	Labelled datasets
Autoencoder	Unsupervised	Medium	No	Statistical structure
LSTM	Unsupervised	Medium	Yes	APTs, temporal threats
CNN	Supervised	Medium	Partial	Payload classification
k-Means / DBSCAN	Unsupervised	High/Med	No	Interpretable clustering

Method Selection Criteria

Choosing the right anomaly detection method depends on your specific requirements regarding data supervision, system scalability, temporal analysis needs, and the type of anomalies you aim to detect.

Isolation Forest

Supervision: Unsupervised

Scalability: High

Temporal: No

Best For: Volumetric anomalies, high-speed monitoring with no labeled data.

OC-SVM

Supervision: Unsupervised

Scalability: Low

Temporal: No

Best For: Identifying compact normal clusters in data.

Random Forest

Supervision: Supervised

Scalability: High

Temporal: No

Best For: Labeled datasets, leveraging feature importance scores.

Autoencoder

Supervision: Unsupervised

Scalability: Medium

Temporal: No

Best For: Detecting deviations from learned statistical data structures.

LSTM

Supervision: Unsupervised

Scalability: Medium

Temporal: Yes

Best For: Advanced Persistent Threats (APTs), temporal threats, slow-and-low exfiltration.

CNN

Supervision: Supervised

Scalability: Medium

Temporal: Partial

Best For: Payload classification and hybrid architectures.

k-Means / DBSCAN

Supervision: Unsupervised

Scalability: High/Medium

Temporal: No

Best For: Interpretable clustering for grouping similar network behaviors.

This guidance helps align your security objectives with the strengths of each machine learning approach for effective anomaly detection.

Deployment Contexts



Enterprise Security

AI-ADS integrated into SIEM platforms. Tiered approach: Isolation Forest for first-pass filtering, LSTM/autoencoder for deep analysis. UEBA extends detection to user accounts.



National Defence

APT campaigns use low-volume, long-horizon tactics. LSTM models temporal behaviour — slow beaconing, lateral movement — invisible to signature tools.



Critical Infrastructure

ICS/SCADA traffic is highly regular, making deviations easy to spot. Autoencoders achieve low false positives. Ukraine power grid attacks (2015–16) show the stakes.

Critical Infrastructure Protection

Protecting essential services from cyber threats is paramount. AI-based anomaly detection systems (AI-ADS) offer robust solutions tailored for the unique characteristics of critical infrastructure.

Highly Regular Traffic Patterns

ICS/SCADA traffic is highly regular, making deviations easy to spot.

Low False Positives with Autoencoders

Autoencoders achieve low false positives in industrial environments.

Lessons from Ukraine

Ukraine power grid attacks (2015–16) demonstrate the critical importance.

Real-time Detection for Vital Systems

Real-time anomaly detection for power systems, water treatment, transportation.

Preventing Catastrophic Failures

Preventing catastrophic failures through early threat detection.

Defense and National Security Applications

AI-based anomaly detection plays a crucial role in national defense and security due to its ability to uncover stealthy and persistent threats that traditional signature-based systems often miss.



APT Campaigns

APT (Advanced Persistent Threat) campaigns use low-volume, long-horizon tactics.



Temporal Analysis

LSTM models temporal behavior effectively.



Stealthy Detection

Detects slow beaconing and lateral movement invisible to signature tools.



State-Sponsored Attacks

Critical for detecting sophisticated state-sponsored attacks.



Integrated Monitoring

Continuous monitoring and threat intelligence integration.

Enterprise Deployment Strategy

1

Integration into SIEM platforms

AI-ADS solutions are integrated into existing Security Information and Event Management (SIEM) platforms to centralize alert correlation and incident response. This ensures a unified view of security posture.

2

Tiered Approach

A tiered approach is employed: **Isolation Forest** is used for first-pass filtering of high-volume traffic, and **LSTM** or **Autoencoder** models are used for deep analysis of suspicious patterns.

3

UEBA Extends Detection

User and Entity Behavior Analytics (UEBA) extends detection capabilities to user accounts and hosts, helping to identify insider threats and compromised credentials by profiling normal behavior.

4

Real-time Monitoring and Alert Generation

The system ensures real-time monitoring and automated alert generation, providing immediate notifications to security teams for prompt investigation and response to detected anomalies.

5

Best Practices for Implementation

Enterprise implementation includes continuous model retraining, performance tuning, and adaptive alert thresholds to stay ahead of evolving threats and reduce false positives.

Implementing these strategies ensures a robust and adaptive anomaly detection system within the enterprise environment.

Mitigation Strategies & Open Challenges

Best Practices

- **Hybrid detection** — combine signature and anomaly-based layers
- **Continuous retraining** — online learning adapts to concept drift in real time
- **Threshold calibration** — tune on local traffic, not public benchmarks
- **Federated learning** — collaborative training without sharing raw data

Open Challenges

Benchmark datasets are outdated; don't reflect modern encrypted traffic.

Deep learning models lack explainability for analyst action.

100+ Gbps throughput requires hardware acceleration.

Models generalize poorly across organizations — transfer learning is an open direction.

Current Research Challenges

Addressing the following challenges is crucial for advancing the efficacy and adoption of AI-based anomaly detection systems in real-world deployments.

1

Outdated Benchmark Datasets

Current benchmark datasets often fail to reflect modern encrypted traffic, which now constitutes the majority of network communications. This disparity leads to models that are ineffective against contemporary threats.

To address this, there's a critical need for developing new, diverse, and representative datasets that incorporate modern traffic patterns and encryption. Focusing on metadata and traffic flow analysis becomes essential when payload content is inaccessible.

2

Lack of Model Explainability

Deep learning models, while powerful, often act as "black boxes," making it difficult for security analysts to understand why a specific anomaly was flagged. This lack of explainability hinders effective investigation, breeds distrust, and slows down incident response.

Research into Explainable AI (XAI) techniques (e.g., LIME, SHAP) is vital to provide actionable insights into model decisions. Developing hybrid approaches that combine complex models with more interpretable components is also a promising direction.

3

High Throughput Requirements

Processing network traffic at speeds exceeding 100 Gbps, common in modern data centers and national infrastructure, poses significant computational challenges for real-time AI-based anomaly detection. Standard CPUs cannot keep pace, risking delayed threat detection.

Solutions involve leveraging specialized hardware accelerators like GPUs, FPGAs, or ASICs. Optimizing AI algorithms for parallel processing and developing more efficient, streamlined model architectures are key to achieving the necessary performance.

4

Poor Generalization Across Organizations

Anomaly detection models trained in one organization often perform poorly when deployed in another due to unique network environments, user behaviors, and threat landscapes. This necessitates extensive retraining and customization, increasing operational costs and deployment time.

Transfer learning and federated learning are promising avenues. Transfer learning allows pre-trained models to be fine-tuned with smaller, organization-specific data, while federated learning enables collaborative model training without sharing sensitive raw data across different entities.

Implementation Best Practices

Hybrid detection

Combine signature and anomaly-based layers for comprehensive coverage.

Continuous retraining

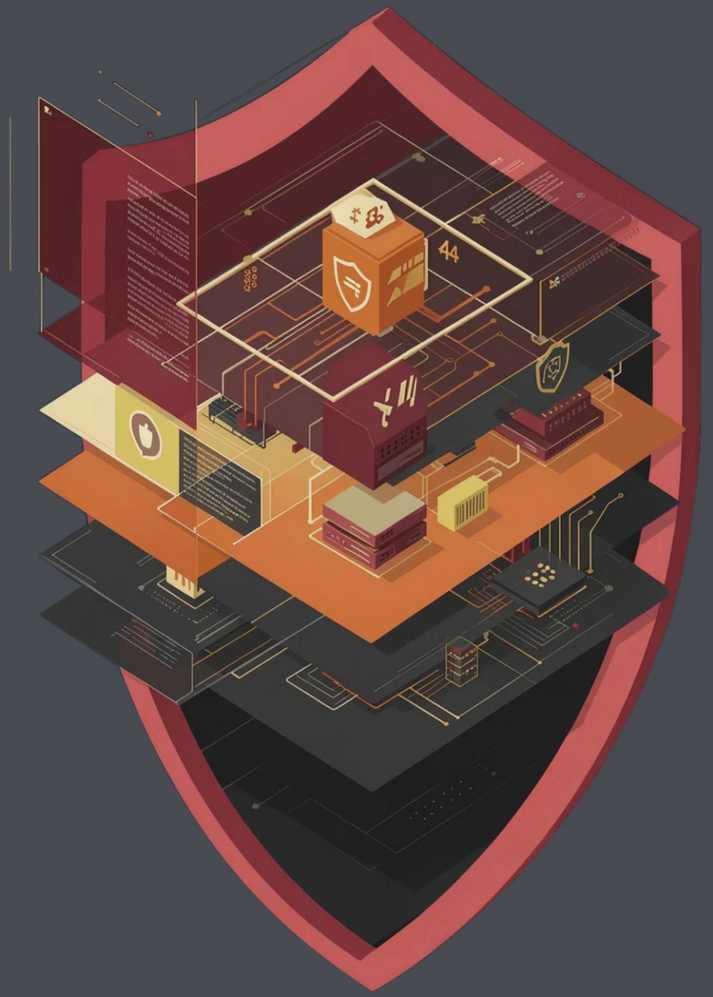
Online learning adapts to concept drift in real time.

Threshold calibration

Tune on local traffic, not public benchmarks.

Federated learning

Collaborative training without sharing raw data.



Conclusion

Signature-based tools are necessary but insufficient. **AI-based anomaly detection** closes the gap against zero-days, covert channels, and slow intrusions.

Isolation Forest

Efficient unsupervised volumetric detection

Autoencoders

Reconstruction error flags unknown attacks

LSTM

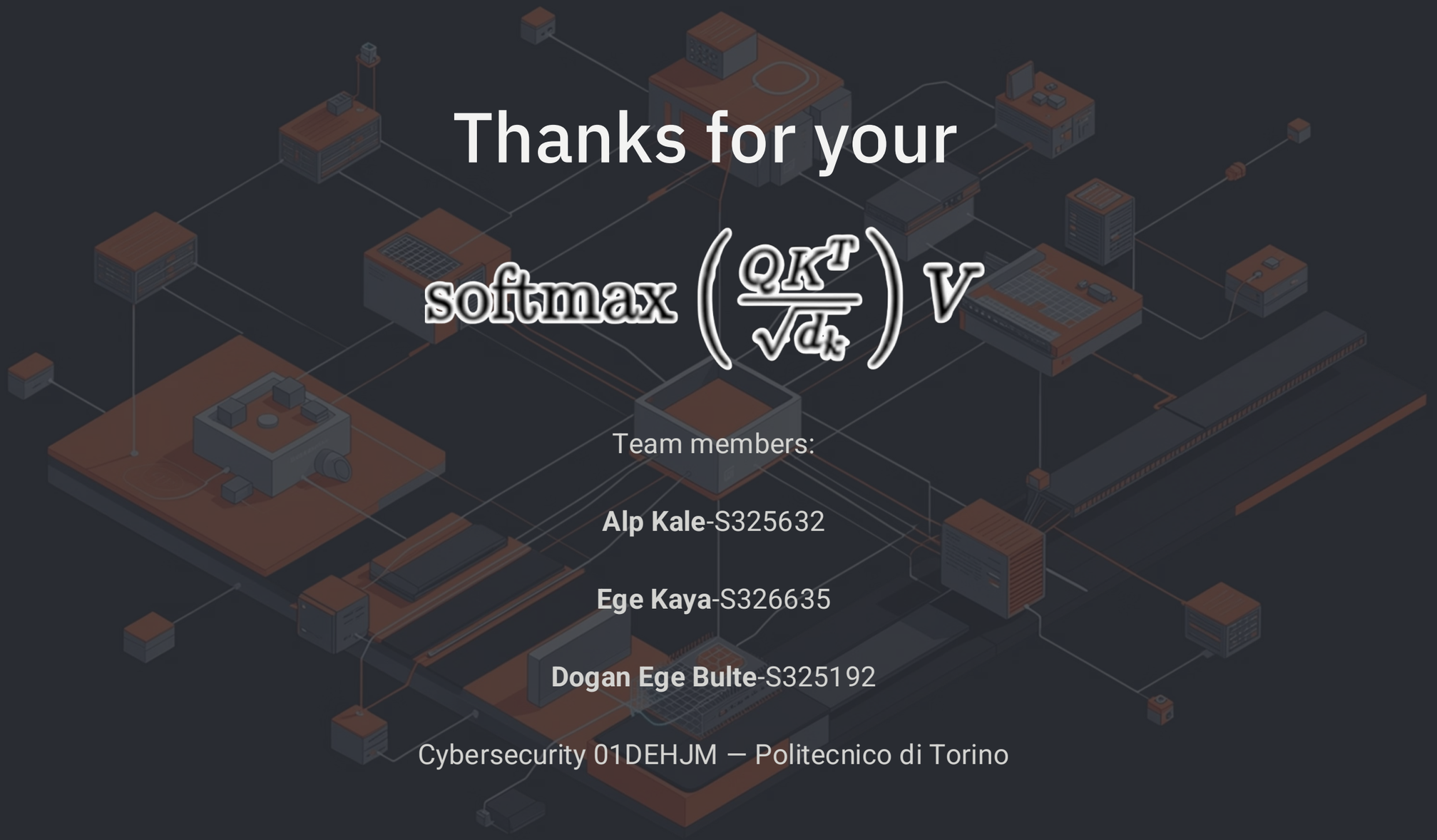
Temporal modeling for APTs and insider threats

Hybrid Architecture

Signature + anomaly + temporal, with continuous retraining

Thanks for your

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



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